Answer Key & Teacher Notes

Session 1 — Customary Unit Conversions · Session 2 — Metric Unit Conversions

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Customary Unit Conversions

TALK PROMPTS

- When you convert larger units (feet) to smaller units (inches), do you multiply or divide? How do you know?
- How many inches are in 1 yard? Use that number to check the gym rope total a second way.

PARTNER TASK

Water Cooler, Party Punch, Gym Rope. One partner converts one fact at a time; the other uses a combined fact (1 yd = 36 in, 1 gal = 16 cups, 1 gal = 8 pt) to check the total. Switch roles for each problem.

WATCH FOR

- Multiplying when changing to a bigger unit (60 in = 720 ft).
- Using a wrong fact, such as 1 ft = 10 in or 1 qt = 2 cups.
- Using the wrong fact to change feet to yards (24 ÷ 12 instead of 24 ÷ 3 for the garden fence).

SESSION 2 · TEACHER NOTES

Metric Unit Conversions

TALK PROMPTS

- Why is converting in the metric system so fast compared to customary units?
- Why did the number get bigger when you changed meters to centimeters?

PARTNER TASK

5K Run, Lab Mass, Lab Ribbon. One partner names the operation and the power of 10 (× 1,000, ÷ 100, …); the other converts and then checks by converting back. Switch roles for each problem.

WATCH FOR

- Dividing when changing to a smaller unit (3 m = 0.03 cm).
- Using the wrong power of 10, such as 1 km = 100 m.
- Dividing amounts in different units (2.4 m ÷ 30 cm) before making the units match.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.RP.A.3d	1	1	1	1	2	2	1	3	2	1	1
S1 · Form B	6.RP.A.3d	1	1	1	1	2	2	1	3	2	1	1
S2 · Form A	6.RP.A.3d	1	1	1	1	2	2	1	3	2	1	1
S2 · Form B	6.RP.A.3d	1	1	1	1	2	2	1	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

customary units: U.S. units like inches, **pounds**, cups and gallons.

convert: Write the same amount with a **different** unit.

conversion factor: A fact that shows two **equal** amounts.

- Big unit to small unit: **multiply**. The number gets bigger.
- Small unit to big unit: **divide**. The number gets smaller.

Example: Quarts to cups: 2 × 4 = **8** cups. Cups to fluid ounces: 8 × 8 = **64** fl oz. Check backward: 64 ÷ 8 = 8 cups, and 8 ÷ 4 = **2** quarts. Answer: 2 quarts = **64** fluid ounces.

YOU TRY

1. a. 9 ft b. 2 lb

2. 7 ÷ 3 = 2 1/3 yards, so buy 3 yards

EXIT TICKET

1. a. 72 in b. 4 gal

2. 2.5 × 16 = 40 oz

3. **A** (multiply)

PRACTICE FORM A

1. a. 48 in b. 15 ft c. 72 oz d. 5 gal e. 3.5 lb f. 12 qt

2. 4 ft = 48 in. Yes, he is exactly 48 inches, so he can ride.

3. a. 2 × 4 = 8 qt b. 8 × 4 = 32 cups

4. a. 8 × 3 = 24 ft b. 24 × 12 = 288 in

5. 2.5 × 5,280 = 13,200 feet

6. 9 × 4 = 36 qt; 36 × 2 = 72 pints, so 72 cups

7. 24 ÷ 4 = 6 gallons

8. a. 24 ÷ 3 = 8 yd b. 8 × $9 = $72 c. Yes. 24 × $3.25 = $78, which is $6 more.

9. Feet are bigger than inches, so the number must get smaller; Sam should divide. 60 ÷ 12 = 5 feet.

10. **C** (60)

11. **D** (3)

Self-check word **INCHES**

PRACTICE FORM B

1. a. 60 in b. 21 ft c. 56 oz d. 7 gal e. 2.5 lb f. 24 qt

2. 5 ft = 60 in. Yes, 60 inches is more than 54 inches.

3. a. 4 × 4 = 16 qt b. 16 × 4 = 64 cups

4. a. 6 × 3 = 18 ft b. 18 × 12 = 216 in

5. 1.5 × 5,280 = 7,920 feet

6. 7 × 4 = 28 qt; 28 × 2 = 56 pints, so 56 cups

7. 36 ÷ 4 = 9 gallons

8. a. 30 ÷ 3 = 10 yd b. 10 × $8 = $80 c. Yes. 30 × $2.90 = $87, which is $7 more.

9. Feet are bigger than inches, so the number must get smaller; Maya should divide. 84 ÷ 12 = 7 feet.

10. **C** (48)

11. **A** (4)

Self-check word **POUNDS**

GUIDED NOTES

metric units: Meters, grams and liters, built on **tens**.

kilo- / centi- / milli-: Kilo means **1,000** times. Centi means one hundredth. Milli means one thousandth.

powers of 10: 10, 100, 1,000. Metric conversions scale place values.

- Big unit to small unit: **multiply** by 10, 100 or 1,000. Small to big: divide.
- Make the units **match** before you divide or compare.

Example: Make the units match: 2.4 × 1,000 = **2,400** g. Divide: 2,400 ÷ 300 = **8**. Check: 8 × 300 = 2,400 g. Answer: You can make **8** portions.

YOU TRY

1. a. 320 cm b. 0.75 kg

2. 15 × 1,000 = 15,000 mL; 15,000 ÷ 750 = 20 jugs

EXIT TICKET

1. a. 6,500 m b. 0.8 m

2. 2 × 1,000 = 2,000 mL; 2,000 ÷ 250 = 8 cups

3. **C** (400 g)

PRACTICE FORM A

1. a. 7,000 m b. 1.5 L c. 3,000 g d. 2.5 m e. 2.45 km f. 0.0025 L

2. 4 L = 4,000 mL; 4,000 ÷ 500 = 8 bottles

3. a. 5 × 1,000 = 5,000 m b. 5,000 − 3,200 = 1,800 m

4. a. 350 × 1,000 = 350,000 mg b. 350 ÷ 1,000 = 0.35 kg

5. a. 2.4 × 100 = 240 cm b. 240 ÷ 30 = 8 pieces

6. 4,800 ÷ 1,000 = 4.8 kg. Yes, 4.8 kg is under 5 kg.

7. 12.75 × 1,000 = 12,750 g

8. a. 8 × 450 = 3,600 m b. 3,600 ÷ 1,000 = 3.6 km c. Sample: kilometers, because 3.6 km is a small, easy number for a long distance.

9. Centimeters are smaller than meters, so the number must get bigger; Daniel should multiply. 3 × 100 = 300 cm.

10. **A** (4,000)

11. **D** (10)

Self-check word **DECIMAL**

PRACTICE FORM B

1. a. 9,000 m b. 2.5 L c. 6,000 g d. 1.75 m e. 3.82 km f. 0.0045 L

2. 6 L = 6,000 mL; 6,000 ÷ 250 = 24 cups

3. a. 6 × 1,000 = 6,000 m b. 6,000 − 2,700 = 3,300 m

4. a. 425 × 1,000 = 425,000 mg b. 425 ÷ 1,000 = 0.425 kg

5. a. 2.4 × 100 = 240 cm b. 240 ÷ 40 = 6 pieces

6. 5,300 ÷ 1,000 = 5.3 kg. No, 5.3 kg is over 5 kg.

7. 9.6 × 1,000 = 9,600 g

8. a. 6 × 750 = 4,500 m b. 4,500 ÷ 1,000 = 4.5 km c. Sample: kilometers, because 4.5 km is a small, easy number for a long distance.

9. Grams are smaller than kilograms, so the number must get bigger; Maya should multiply. 5 × 1,000 = 5,000 g.

10. **B** (2.5)

11. **C** (3,000)

Self-check word **METRICS**

SHEET 1 · CUSTOMARY UNIT CONVERSIONS

1. a. 96 in b. 7 yd c. 80 oz d. 3 gal e. 14 cups f. 2.5 lb

2. a. < (3 ft = 36 in) b. < (5 pt = 2.5 qt) c. > (2 lb = 32 oz) d. = (3 yd = 9 ft)

3. 3 gal = 12 qt; 20 qt = 5 gal; 7.5 gal = 30 qt

4. 3 gal = 12 qt = 24 pt = 48 cups

5. No. 2 yd = 2 × 3 = 6 ft = 6 × 12 = 72 in. 70 in is less than 72 in, so she is 2 inches short.

SHEET 2 · METRIC UNIT CONVERSIONS

1. a. 3,500 m b. 0.85 m c. 6,000 mL d. 0.45 kg

2. a. 18 × 1,000 = 18,000 mL b. 18,000 ÷ 300 = 60 cups

3. a. 4 × 800 = 3,200 m b. 3,200 ÷ 1,000 = 3.2 km

4. a. 3 × 400 = 1,200 g b. No. 1.5 kg = 1,500 g; 1,500 − 1,200 = 300 g short

5. He divided by 1,000 instead of multiplying. Kilograms to grams is large to small, so multiply: 4.5 × 1,000 = 4,500 g.

6. **B** (160 mm)

7. **C** (600 mL)

October 2026 conceptual additions — teacher key

Session 1 · Customary Unit Conversions

8 ÷ 3 = 2 2/3 yards; buy 3 whole yards. Two yards is 6 feet, which is insufficient. The purchase is 9 feet, leaving 1 foot.

Extension: They measure different quantities: mass/weight versus volume. A conversion must connect units of the same measured quantity or provide a context-specific relationship.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Metric Unit Conversions

1 L = 1000 mL; 0.5 L = 500 mL; 1.5 L = 1500 mL. 1500 ÷ 250 = 6 cups. Smaller units produce a larger numerical measure of the same quantity.

Extension: 1 × 1000 + 0.5 × 1000 = 1000 + 500 = 1500; each part is scaled by the same factor.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.